Bridge Graphical Models: Coupling, Projection, and Gauge in Generative Modeling

Anonymous Author(s)
Affiliation
Address
email

Abstract

Diffusion models, score-based SDEs, rectified flows, flow matching, Schrödinger bridges, and Poisson-flow generative models are usually described as separate paradigms. We propose *Bridge Graphical Models* (BGMs), a path-space latent-variable view that decomposes these methods into four choices: an endpoint coupling, a conditional bridge law, a Markovian projection from non-causal bridge paths to causal dynamics, and a probability-current gauge that represents the same marginal evolution as an ODE, SDE, or augmented field flow. Beyond unifying notation, we contribute two diagnostics and one formalization. First, the *Markovization gap* is an irreducible conditional-variance loss measuring how much bridge information is lost under Markov compression. Second, a stability bound connects low projection error to terminal distribution error. Third, Poisson and electrostatic models become field-line bridge kernels induced by hitting maps of augmented potential fields. Synthetic diagnostics and a no-download real-image latent benchmark show that changing only the endpoint coupling changes both the Markovization gap and downstream velocity-regression difficulty. On a two-dimensional mixture, minibatch OT reduces the gap by roughly two orders of magnitude; on PCA latents of an LFW face subset, it roughly halves the gap and reduces fixed-model velocity error by about 45%. The paper is intended as a concept-and-feasibility contribution: it gives a graphical-model language, proof-level diagnostics, and a minimal empirical testbed for bridge design.

1 Introduction

Continuous-time generative modeling now has several successful but partly disconnected languages. Diffusion models reverse noising Markov chains and are trained through variational bounds [4, 7, 18]. Score-based SDEs give reverse-time SDE samplers and probability-flow ODEs [19]. Flow matching and rectified flow learn ODE velocities by regressing conditional paths [11, 12, 20]. Schrödinger bridge methods replace deterministic transport with entropy-regularized stochastic transport [15, 21]. Poisson-flow and electrostatic field-matching methods embed data into an augmented space and follow field lines generated by charges or interactions [8, 13, 16, 24, 25].

We argue that the common object is not a score, velocity, SDE, ODE, or electric field. It is a latent path graphical model: couple a base endpoint with a data endpoint, sample a conditional path bridge between them, project this non-causal path law into a causal Markov decoder, and choose a gauge for representing the resulting probability current. Under this view, diffusion and rectified flow are different bridge/gauge choices, while Poisson-style field methods are augmented-space bridge geometries rather than exceptions.

Contributions. (i) We define BGMs as endpoint couplings plus bridge kernels plus Markov decoders. (ii) We introduce the Markovization gap, an irreducible bridge-selection quantity. (iii)

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Table 1: Positioning relative to active unification directions.

Framework	Main primitive	BGM addition
Stochastic interpolants	interpolating process and current	endpoint inference coupling and VAE-like path graph
DFM / bridge matching	coupling, bridge, Markov projection	projection gap as bridge-selection diagnostic
Latent stochastic interpolants	latent ELBO and encoder/decoder	bridge law and gauge as independent choices
PFGM / EFM / IFM	augmented electrostatic/interaction field	hitting-map bridge kernels and caustic gap

We prove a simple stability bound showing how projection error controls terminal Wasserstein error. (iv) We formalize Poisson/electrostatic flows as field-line bridge kernels with a caustic gap. (v) We provide reproducible diagnostics on both a two-dimensional toy problem and a real-image latent benchmark showing that coupling choice changes Markovization difficulty and velocity-regression error.

2 Relation to existing unifications

Stochastic interpolants already unify deterministic flows and stochastic diffusions through interpolating processes and probability currents [1, 2]. Diffusion Flow Matching (DFM) describes generation as choosing a coupling and a deterministic or stochastic bridge, then learning a Markovian projection, and gives KL guarantees for Brownian choices [3]. Unified bridge algorithms relate flow matching, OT flow matching, Schrödinger bridge flow matching, and DSBM [6]. Latent stochastic interpolants add a learned encoder/decoder and continuous-time ELBO [17]. Field-matching duality studies when interaction-field models and conditional flow matching describe equivalent dynamics [16].

Table 1 summarizes the distinction. The present paper does not replace these frameworks; it isolates four graphical-model primitives and adds diagnostics for when a bridge is easy to compress into a Markov decoder.

3 Bridge graphical models

Let $\mathcal{M}$ be a state space, often $\mathbb{R}^d$ or an augmented space $\mathbb{R}^d \times \mathbb{R}^D$, and let $\mathcal{X} = C([0,1], \mathcal{M})$ with coordinate process X_t. Let π_0 be a simple base distribution and $\pi_1 = \pi_{\mathrm{data}}$ the data distribution.

Definition 1 (Endpoint coupling)**.** *An endpoint coupling is* $\Gamma(\mathrm{d}z, \mathrm{d}x) \in \Pi(\pi_0, \pi_1)$. *Examples include independent coupling, minibatch OT, entropic OT, paired-data coupling, or a learned encoder coupling* $\Gamma_\phi(\mathrm{d}z, \mathrm{d}x) = q_\phi(\mathrm{d}z \mid x)\pi_{\mathrm{data}}(\mathrm{d}x)$.

Definition 2 (Bridge kernel and teacher path law)**.** *For each endpoint pair* (z, x)*, a bridge kernel is a path-space probability measure* $\mathsf{R}_\lambda^{z,x}(\mathrm{d}X_{0:1})$ *with* $X_0 = z$ *and* $X_1 = x$*. The aggregated teacher path law is*

$$\mathsf{Q}_{\Gamma,\lambda}(\mathrm{d}X_{0:1}) = \int \mathsf{R}_\lambda^{z,x}(\mathrm{d}X_{0:1})\, \Gamma(\mathrm{d}z, \mathrm{d}x). \tag{3.1}$$

Definition 3 (Bridge graphical model)**.** *A BGM is a tuple* $(\Gamma_\phi, \mathsf{R}_\lambda, \mathsf{P}_\theta, p_\psi)$*: an endpoint inference coupling, a bridge kernel, a causal Markov path law initialized from* π_0*, and an optional observation decoder. Formally,*

$$\log p_{\theta,\psi}(x) \geq \mathbb{E}_{\mathsf{Q}_\phi(\mathrm{d}Y_{0:1}|x)} \left[\log p_\psi(x \mid Y_1) - \log \frac{\mathrm{d}\mathsf{Q}_\phi(\cdot \mid x)}{\mathrm{d}\mathsf{P}_\theta}(Y_{0:1}) \right], \tag{3.2}$$

whenever the inference path law is absolutely continuous with respect to the generative path law.

Equation (3.2) is a continuous-depth VAE: the latent variable is the whole path. If two path laws share diffusion covariance and differ only in drift, Girsanov's theorem reduces the path KL to an integrated drift-matching cost. Figure 1 shows the resulting computation graph.

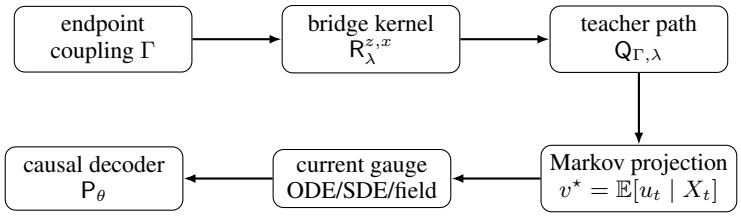

Figure 1: The BGM decomposition. Existing models usually fix several boxes implicitly; the framework makes coupling, bridge, projection, and gauge explicit and therefore comparable.

4 Markovian projection and gap

Assume Q has a local velocity target u_t in the deterministic weak sense. A causal ODE decoder $\mathrm{d}Y_t = v_\theta(Y_t, t)\mathrm{d}t$ cannot condition on both endpoints; it only sees (Y_t, t). This causalization is the population operation behind flow matching.

Definition 4 (Markovization gap)**.**

$$\mathfrak{G}(\mathsf{Q}) = \inf_v \int_0^1 \mathbb{E}_{\mathsf{Q}} \|u_t - v(X_t, t)\|^2 \mathrm{d}t. \tag{4.1}$$

Proposition 1 (Projection identity)**.** *The population minimizer of the matching loss is*

$$v^\star(x, t) = \mathbb{E}_{\mathsf{Q}}[u_t \mid X_t = x], \tag{4.2}$$

and

$$\mathfrak{G}(\mathsf{Q}) = \int_0^1 \mathbb{E}_{\mathsf{Q}} \operatorname{Tr} \operatorname{Var}(u_t \mid X_t)\mathrm{d}t. \tag{4.3}$$

The gap depends on the bridge/coupling rather than on neural capacity. Low gap means that the chosen non-causal bridge is almost Markov after observing X_t; high gap means that multiple endpoint configurations induce conflicting velocities at the same location and time.

Theorem 1 (Terminal stability from projection error)**.** *Let ρ_t be transported by $v^\star$, and let ρ_t^θ be transported by v_θ from the same initial law. If $v_\theta(\cdot, t)$ is L-Lipschitz and*

$$\epsilon^2 = \int_0^1 \mathbb{E}_{X_t \sim \rho_t} \|v_\theta(X_t, t) - v^\star(X_t, t)\|^2 \mathrm{d}t, \tag{4.4}$$

then

$$W_2(\rho_1^\theta, \rho_1) \le e^L \epsilon. \tag{4.5}$$

Moreover the excess matching loss equals the squared projection error: $\mathcal{L}_{\mathrm{FM}}(v_\theta) - \mathfrak{G}(\mathsf{Q}) = \epsilon^2$.

Thus $\mathfrak{G}(\mathsf{Q})$ is a floor: once a bridge is chosen, the part of the regression loss above this floor controls marginal error, while the floor itself measures bridge compressibility.

5 Probability-current gauge

Let ρ_t solve $\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$ with current $\boldsymbol{J}_t = \rho_t v_t$. For an SDE $\mathrm{d}X_t = b_t(X_t)\mathrm{d}t + \sigma_t(X_t)\mathrm{d}B_t$ and $a_t = \sigma_t \sigma_t^\top$, the Fokker–Planck current is

$$J_{t,i} = \rho_t b_{t,i} - \frac{1}{2} \sum_j \partial_j (a_{t,ij} \rho_t). \tag{5.1}$$

Theorem 2 (Probability-current gauge)**.** *Given ρ_t, v_t as above and a smooth $a_t(x) \succeq 0$, define*

$$b_i^{(a)}(x, t) = v_i(x, t) + \frac{1}{2\rho_t(x)} \sum_j \partial_j (a_{ij}(x, t)\rho_t(x)). \tag{5.2}$$

Then the SDE with drift $b^{(a)}$ and diffusion matrix a has the same marginal path ρ_t whenever the Fokker–Planck equation is well posed. If $a_t = g(t)^2 I$, then $b^{(a)} = v_t + \frac{g(t)^2}{2} \nabla \log \rho_t$.

The marginal evolution is determined by the current. The gauge determines whether this current is represented deterministically ($a = 0$), stochastically ($a \neq 0$ plus score correction), or in an augmented field coordinate system.

Table 2: Several generative model families as BGM choices.

Family	Coupling / bridge	Learned object	Gauge
DDPM / score SDE	fixed Gaussian noising bridge	score or denoiser	SDE / path ELBO
Flow matching	conditional probability path	velocity field	ODE
Rectified flow	straight bridge $X_t = (1-t)Z + tX$	$\mathbb{E}[X - Z \mid X_t]$	ODE
Brownian / DFM	Brownian bridge over endpoints	projected drift	SDE
Schrödinger bridge	KL-optimal Brownian-reference path	forward/backward drifts	entropy regularized
PFGM / EFM / IFM	hitting-map field-line bridge	field or interaction field	augmented field

6 Field-line bridge kernels

Let $\Omega \subset \mathbb{R}^m$ have source and target hypersurfaces S_0, S_1. Let $\Phi : \Omega \to \mathbb{R}$ be a potential and $\boldsymbol{F} = -\nabla\Phi$ its field. Let $\varphi_s(z)$ solve $\dot{\varphi}_s = \boldsymbol{F}(\varphi_s)$, $\varphi_0 = z$. Define the hitting time and hitting map

$$\tau(z) = \inf\{s > 0 : \varphi_s(z) \in S_1\}, \qquad T_\Phi(z) = \varphi_{\tau(z)}(z). \tag{6.1}$$

After choosing a clock $r_z : [0,1] \to [0, \tau(z)]$, set $\gamma_z(t) = \varphi_{r_z(t)}(z)$ and $u_t(z) = \partial_t \gamma_z(t)$.

Definition 5 (Field-line bridge)**.** *Given a source measure μ_0 on S_0, the field-induced coupling and bridge are*

$$\Gamma_\Phi = (\mathrm{Id}, T_\Phi)_\# \mu_0, \qquad \mathsf{R}_\Phi^{z,x} = \delta_{\gamma_z}, \quad x = T_\Phi(z). \tag{6.2}$$

This makes PFGM, PFGM++, electrostatic field matching, and interaction-field matching bridge geometries in augmented space. A regularized version uses $\mathrm{d}Y_s = \boldsymbol{F}(Y_s)\mathrm{d}s + \sqrt{2\varepsilon}\mathrm{d}B_s$ and the Doob bridge $\mathcal{L}(Y_{0:1} \mid Y_0 = z, Y_1 = x)$.

Theorem 3 (Caustic gap)**.** *For $X_t = \gamma_Z(t)$, $Z \sim \mu_0$, the marginals $\rho_t = (\gamma_t)_\# \mu_0$ satisfy a weak continuity equation with vector-valued current $\boldsymbol{J}_t = (\gamma_t)_\#(u_t \mu_0)$. The Markov velocity $v^\star(y,t) = \mathbb{E}[u_t(Z) \mid \gamma_Z(t) = y]$ realizes the same one-time marginals, and the irreducible field-to-flow compression loss is*

$$\mathfrak{C}_\Phi = \int_0^1 \mathbb{E}\,\mathrm{Tr}\,\mathrm{Var}(u_t(Z) \mid \gamma_Z(t))\mathrm{d}t. \tag{6.3}$$

It is zero exactly when the field-line velocity is a function of the current point-time a.s.

7 Instantiations

For Gaussian paths $X_t = \alpha_t X + \sigma_t \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$, Tweedie conditioning gives $\mathbb{E}[\varepsilon \mid X_t = x] = -\sigma_t \nabla \log \rho_t(x)$, and

$$v^\star(x,t) = \frac{\dot{\alpha}_t}{\alpha_t} x + \left(\frac{\dot{\alpha}_t \sigma_t^2}{\alpha_t} - \dot{\sigma}_t \sigma_t \right) \nabla \log \rho_t(x). \tag{7.1}$$

Thus score, noise, denoiser, and velocity prediction are schedule-dependent reparameterizations of the same conditional information.

This also clarifies how model families differ. Diffusion fixes a noisy encoder bridge and learns a reverse Markov decoder; rectified flow fixes a deterministic bridge and learns its causal projection; Schrödinger methods optimize the bridge under an entropic reference; field methods define the bridge indirectly by a potential and a hitting map. The same paper can therefore ask modular questions: should the endpoint coupling be learned, should the bridge be Brownian or deterministic, and should sampling use an ODE or SDE gauge? The Markovization gap answers the second question only after the first has been fixed, which is why coupling and bridge geometry should be evaluated together rather than separately.

8 Diagnostics and latent real-data benchmark

We test the operational prediction that coupling choices with lower Markovization gap should impose an easier velocity-regression problem under the same bridge and learner. The first test is synthetic and

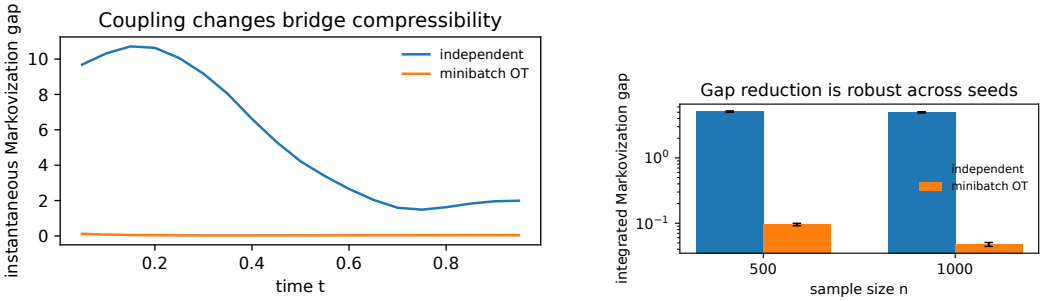

Figure 2: Synthetic Markovization-gap diagnostics. Left: one run with $n = 1000$ gives integrated gap 4.877 for independent pairing and 0.038 for minibatch OT. Right: five-seed replication at $n \in \{500, 1000\}$; OT consistently lowers the irreducible compression floor by about two orders of magnitude.

Table 3: Real-image latent benchmark on the LFW subset, mean $\pm$ standard deviation over three seeds. Lower is better.

Coupling	gap AUC	eval velocity MSE	latent MMD^2
independent	23.701 ± 0.880	23.511 ± 1.385	0.02780 ± 0.00649
minibatch OT	11.871 ± 0.201	12.877 ± 0.078	0.02744 ± 0.00686

isolates the diagnostic. Base samples are drawn from $\mathcal{N}(0, I_2)$; target samples are an eight-Gaussian ring. The bridge is fixed to $X_t = (1-t)Z + tX$, $u_t = X - Z$. We compare independent endpoint pairing to minibatch OT assignment. At 19 times in $[0.05, 0.95]$, $\mathbb{E}\,\mathrm{Tr}\,\mathrm{Var}(u_t \mid X_t)$ is estimated by a nearest-neighbor conditional-variance estimator.

We then run a small real-image latent benchmark. We use the no-download LFW face subset bundled with scikit-image, embed images into 16-dimensional PCA-whitened latents, use the same straight bridge, and compare the same two couplings over three seeds. For each coupling we estimate the gap and fit a fixed pure-NumPy ridge velocity model using features $(X_t, t, X_t t, \sin \pi t, \cos \pi t)$. This is not a state-of-the-art image benchmark; it is a controlled test of whether the gap predicts the difficulty of Markov compression on real data.

The LFW result supports the intended use of $\mathfrak{G}$: minibatch OT roughly halves the estimated gap and reduces fixed-model velocity error by about 45% under the same bridge. Latent MMD is nearly unchanged in this tiny benchmark, so we do not claim improved sample quality; the evidence is for bridge compressibility and training difficulty. The supplementary repository also includes scripts for MNIST, Fashion-MNIST, and CIFAR-10 latent runs.

9 Design consequences

The decomposition suggests three concrete design rules. First, choose or learn the endpoint coupling as an inference problem: independent coupling asks the generator to discover semantic alignment, while OT or encoder-induced coupling moves part of that alignment into the bridge posterior. Second, evaluate bridges by their compressibility, not only by visual smoothness; a bridge with low curvature may still have high Markovization gap if many hidden endpoint pairings collide at the same state. Third, treat the sampler as a gauge choice after the current has been specified. ODE gauges favor likelihood and deterministic transport; SDE gauges can improve exploration or robustness but require score/divergence corrections; augmented field gauges may preserve path information that a data-space Markov velocity loses at caustics.

10 Limitations and outlook

The current evidence is intentionally small: two-dimensional synthetic diagnostics, no image-scale training, and no learned bridge family. The stability theorem assumes Lipschitz velocities and mea-

sures error under the teacher path; high-dimensional estimation of $\mathfrak{G}$ will require better conditional-variance estimators. The field-line construction assumes well-defined hitting maps away from singular charge sets; small-noise Doob bridges are a regularization rather than a complete numerical method. The next step is to test whether lower Markovization gap predicts faster convergence, fewer solver steps, or better sample quality at scale, and to learn Γ_ϕ and R_λ jointly under the path ELBO.

References

References

[1] Michael S. Albergo and Eric Vanden-Eijnden. Building normalizing flows with stochastic interpolants. In *International Conference on Learning Representations*, 2023.

[2] Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *arXiv preprint*, 2023.

[3] Marta Gentiloni Silveri, Giovanni Conforti, and Alain Durmus. Theoretical guarantees in KL for diffusion flow matching. In *Advances in Neural Information Processing Systems*, 2024.

[4] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.

[5] Gary B. Huang, Manu Ramesh, Tamara Berg, and Erik Learned-Miller. Labeled faces in the wild: A database for studying face recognition in unconstrained environments. Technical Report 07-49, University of Massachusetts, Amherst, 2007.

[6] Minyoung Kim. A unified framework for diffusion bridge problems: Flow matching and Schrödinger matching into one. *arXiv preprint*, 2025.

[7] Diederik P. Kingma, Tim Salimans, Ben Poole, and Jonathan Ho. Variational diffusion models. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[8] Alexander Kolesov, Stepan I. Manukhov, Vladimir V. Palyulin, and Alexander Korotin. Field matching: an electrostatic paradigm to generate and transfer data. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025.

[9] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

[10] Yann LeCun, Corinna Cortes, and Christopher J. C. Burges. The MNIST Database of Handwritten Digits, 1998. Online dataset.

[11] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.

[12] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations*, 2023.

[13] Stepan I. Manukhov, Alexander Kolesov, Vladimir V. Palyulin, and Alexander Korotin. Interaction field matching: Overcoming limitations of electrostatic models. *arXiv preprint*, 2025.

[14] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[15] Yuyang Shi, Valentin De Bortoli, Andrew Campbell, and Arnaud Doucet. Diffusion Schrödinger bridge matching. In *Advances in Neural Information Processing Systems*, 2023.

[16] Daniil Shlenskii, Alexander Varlamov, Nazar Buzun, and Alexander Korotin. Unlocking the duality between flow and field matching. *arXiv preprint*, 2026.

[17] Saurabh Singh and Dmitry Lagun. Latent stochastic interpolants. *arXiv preprint*, 2025.

[18] Jascha Sohl-Dickstein, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *Proceedings of the 32nd International Conference on Machine Learning*, pages 2256–2265, 2015.

[19] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

[20] Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Huguet, Yanlei Zhang, Jarrid Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024.

[21] Alexander Tong, Nikolay Malkin, Kilian Fatras, Lazar Atanackovic, Yanlei Zhang, Guillaume Huguet, Guy Wolf, and Yoshua Bengio. Simulation-free Schrödinger bridges via score and flow matching. In *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, 2024.

[22] Stéfan van der Walt, Johannes L. Schönberger, Juan Nunez-Iglesias, François Boulogne, Joshua D. Warner, Neil Yager, Emmanuelle Gouillart, Tony Yu, and the scikit-image contributors. scikit-image: image processing in Python. *PeerJ*, 2:e453, 2014.

[23] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.

[24] Yilun Xu, Ziming Liu, Max Tegmark, and Tommi Jaakkola. Poisson flow generative models. In *Advances in Neural Information Processing Systems*, 2022.

[25] Yilun Xu, Ziming Liu, Yonglong Tian, Shangyuan Tong, Max Tegmark, and Tommi Jaakkola. PFGM++: Unlocking the potential of physics-inspired generative models. In *Proceedings of the 40th International Conference on Machine Learning*, 2023.

A Formal path-space setup

Let $(\mathcal{M}, \mathcal{B})$ be a Polish state space and $\mathcal{X} = C([0,1], \mathcal{M})$ with the Borel sigma-algebra induced by the uniform topology. The coordinate process is $X_t(\omega) = \omega(t)$ and the endpoint map is $e = (X_0, X_1)$. A bridge kernel is a probability kernel $(z, x) \mapsto \mathsf{R}^{z,x}$ such that $\mathsf{R}^{z,x}(e^{-1}(z, x)) = 1$. If Q is any path measure with endpoint law $\Gamma = e_{\#}\mathsf{Q}$, then by disintegration on Polish spaces there exists a bridge kernel satisfying

$$\mathsf{Q}(\mathrm{d}\omega) = \int \mathsf{R}^{z,x}(\mathrm{d}\omega)\Gamma(\mathrm{d}z, \mathrm{d}x). \tag{A.1}$$

Conversely, any coupling Γ and bridge kernel define the teacher path law in Eq. (3.1). This justifies treating the coupling and bridge as separate graphical-model components.

For deterministic bridges, assume paths are absolutely continuous and that $\int_0^1 \mathbb{E}\|u_t\|^2 \mathrm{d}t < \infty$, where $u_t = \partial_t X_t$. Then the vector-valued measure $\boldsymbol{J}_t(A) = \mathbb{E}[u_t \mathbf{1}\{X_t \in A\}]$ is finite for almost every t. For stochastic bridges, the analogous object is the weak probability current; when an Itô drift β_t exists, the Markov projection replaces u_t by β_t.

B Path-space ELBO and Girsanov reduction

Suppose the generative path law P_θ starts from π_0 and evolves by

$$\mathrm{d}Y_t = b_\theta(Y_t, t)\mathrm{d}t + \sigma_t(Y_t)\mathrm{d}B_t, \tag{B.1}$$

and the inference bridge conditional on x has drift $\beta_\phi(Y_{0:t}, t, x)$ with the same diffusion covariance $a_t = \sigma_t \sigma_t^\top$. If Novikov-type integrability holds and the two path laws share support, Girsanov's theorem gives

$$\mathrm{KL}(\mathsf{Q}_\phi(\cdot \mid x)\|\mathsf{P}_\theta) = \frac{1}{2}\mathbb{E}_{\mathsf{Q}_\phi}\int_0^1 \|\sigma_t^{-1}(\beta_\phi - b_\theta(Y_t, t))\|^2 \mathrm{d}t + \mathrm{KL}(q_\phi(Y_0 \mid x)\|\pi_0), \tag{B.2}$$

up to endpoint/decoder terms. Thus a path-space VAE objective reduces to drift matching in a shared-covariance gauge. In the zero-noise limit, the Radon–Nikodym derivative becomes singular, but the corresponding regression objective persists as deterministic flow matching.

C Proof of Proposition 1

For a fixed time t, let $\mathcal{H}_t$ be the closed subspace of $L^2(\mathsf{Q})$ consisting of functions measurable with respect to $\sigma(X_t)$. The conditional expectation $\mathbb{E}[u_t \mid X_t]$ is the orthogonal projection of u_t onto $\mathcal{H}_t$. For any measurable v,

$$\mathbb{E}\|u_t - v(X_t, t)\|^2 = \mathbb{E}\|u_t - \mathbb{E}[u_t \mid X_t] + \mathbb{E}[u_t \mid X_t] - v(X_t, t)\|^2 \tag{C.1}$$

$$= \mathbb{E}\,\mathrm{Tr}\,\mathrm{Var}(u_t \mid X_t) + \mathbb{E}\|\mathbb{E}[u_t \mid X_t] - v(X_t, t)\|^2, \tag{C.2}$$

where the cross term is zero because $v(X_t, t)$ is $\sigma(X_t)$-measurable. Integrating over t proves the minimizer and Eq. (4.3). Uniqueness holds up to ρ_t-null sets for almost every t.

For an Itô bridge $\mathrm{d}X_t = \beta_t \mathrm{d}t + \sigma_t \mathrm{d}B_t$ with adapted non-Markov drift, the Hilbert-space projection gives $b^\star(x, t) = \mathbb{E}[\beta_t \mid X_t = x]$. Under the hypotheses of Markovian mimicking theorems, this projected Markov diffusion has the same one-time marginals as the original semimartingale. The BGM framework uses this projection identity as the population target behind stochastic bridge matching.

D Proof of Theorem 1 and bridge comparison

Couple the two flows by the same initial random variable $X_0 = Y_0 \sim \pi_0$. Let X_t solve $\dot{X}_t = v^\star(X_t, t)$ and Y_t solve $\dot{Y}_t = v_\theta(Y_t, t)$. Then

$$\|Y_t - X_t\| \leq \int_0^t \|v_\theta(Y_s, s) - v^\star(X_s, s)\| \mathrm{d}s \tag{D.1}$$

$$\leq \int_0^t L\|Y_s - X_s\| \mathrm{d}s + \int_0^t \|v_\theta(X_s, s) - v^\star(X_s, s)\| \mathrm{d}s. \tag{D.2}$$

Gronwall's inequality gives

$$\|Y_1 - X_1\| \leq e^L \int_0^1 \|v_\theta(X_s, s) - v^\star(X_s, s)\| \mathrm{d}s. \tag{D.3}$$

Taking L^2 norms and applying Cauchy–Schwarz gives Eq. (4.5). Since W_2 is bounded by the cost of any coupling, the bound follows. The excess matching identity is obtained by subtracting $\mathfrak{G}(\mathsf{Q})$ from the orthogonal decomposition in Appendix C.

Corollary 1 (Bridge comparison)**.** *Consider two teacher laws Q^A and Q^B with the same endpoints and with Markov projections $v_A^\star, v_B^\star$. If a function class can approximate both projections to comparable error and $\mathfrak{G}(\mathsf{Q}^A) < \mathfrak{G}(\mathsf{Q}^B)$, then bridge A has a lower irreducible regression floor. Combined with Theorem 1, this means that lowering the Markovization gap is a sufficient design heuristic for reducing the attainable terminal-error bound when approximation and optimization errors are controlled.*

E Proof of Theorem 2

The Fokker–Planck equation for $\mathrm{d}X_t = b_t \mathrm{d}t + \sigma_t \mathrm{d}B_t$ is

$$\partial_t \rho_t = -\sum_i \partial_i (b_i \rho_t) + \frac{1}{2} \sum_{i,j} \partial_i \partial_j (a_{ij} \rho_t). \tag{E.1}$$

Writing this as $\partial_t \rho_t + \nabla \cdot \boldsymbol{J}_t = 0$ gives

$$J_{t,i} = \rho_t b_{t,i} - \frac{1}{2} \sum_j \partial_j (a_{ij} \rho_t). \tag{E.2}$$

Substituting Eq. (5.2) yields $J_{t,i} = \rho_t v_i$. Therefore the SDE and ODE have the same current and the same marginal path by uniqueness of the Fokker–Planck solution. For $a_t = g(t)^2 I$, $\partial_i (g^2 \rho_t) = g^2 \partial_i \rho_t$, giving the score correction.

A useful equivalent statement is that the same continuity equation can be decomposed into drift and diffusion in infinitely many ways. The score is not an extra target independent of velocity; it is the term that compensates for adding diffusion while preserving the current.

F Field-line bridges and caustics

Assume $\boldsymbol{F}$ is locally Lipschitz away from the charge set, the flow exists until S_1, and the hitting time $\tau(z)$ is finite μ_0-a.s. Transversality of the field to S_1 makes T_Φ measurable and smooth on regular

regions. For any smooth compactly supported test function f,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int f(y)\rho_t(\mathrm{d}y) = \frac{\mathrm{d}}{\mathrm{d}t}\int f(\gamma_z(t))\mu_0(\mathrm{d}z) \tag{F.1}$$

$$= \int \nabla f(\gamma_z(t)) \cdot u_t(z)\mu_0(\mathrm{d}z) \tag{F.2}$$

$$= \int \nabla f(y) \cdot \boldsymbol{J}_t(\mathrm{d}y), \tag{F.3}$$

where $\boldsymbol{J}_t = (\gamma_t)_\#(u_t\mu_0)$ is the vector-valued pushforward measure. This is the weak form of the continuity equation. Disintegrating μ_0 with respect to γ_t gives

$$\mu_0(\mathrm{d}z) = \mu_t(\mathrm{d}z \mid y)\rho_t(\mathrm{d}y), \qquad y = \gamma_t(z), \tag{F.4}$$

and therefore $\boldsymbol{J}_t(\mathrm{d}y) = \rho_t(\mathrm{d}y)\int u_t(z)\mu_t(\mathrm{d}z \mid y) = \rho_t(\mathrm{d}y)v^\star(y,t)$. The caustic-gap identity is Proposition 1 applied to $X_t = \gamma_Z(t)$ and $u_t = u_t(Z)$.

If γ_t is injective μ_0-a.s., then conditioning on $\gamma_Z(t) = y$ recovers Z and the conditional variance is zero. If multiple field lines pass through the same point-time with distinct velocities, the conditional variance is positive. This is why the caustic gap measures information retained by the augmented path representation but lost by a data-space Markov velocity.

G Gaussian bridge reparameterization

Let $X_t = \alpha_t X + \sigma_t\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$ independent of $X \sim \pi_{\mathrm{data}}$, and assume $\alpha_t \neq 0$. The conditional density is $p_t(x \mid X) \propto \exp(-\|x - \alpha_t X\|^2/(2\sigma_t^2))$. Differentiating the marginal density gives

$$\nabla \log \rho_t(x) = \mathbb{E}\left[-\frac{x - \alpha_t X}{\sigma_t^2} \mid X_t = x\right]. \tag{G.1}$$

Since $x - \alpha_t X = \sigma_t\varepsilon$ under the conditional path,

$$\mathbb{E}[\varepsilon \mid X_t = x] = -\sigma_t \nabla \log \rho_t(x). \tag{G.2}$$

Also $X = (X_t - \sigma_t\varepsilon)/\alpha_t$, so

$$\mathbb{E}[\dot{X}_t \mid X_t = x] = \mathbb{E}[\dot{\alpha}_t X + \dot{\sigma}_t\varepsilon \mid X_t = x] \tag{G.3}$$

$$= \frac{\dot{\alpha}_t}{\alpha_t}x + \left(\dot{\sigma}_t - \frac{\dot{\alpha}_t\sigma_t}{\alpha_t}\right)\mathbb{E}[\varepsilon \mid X_t = x] \tag{G.4}$$

$$= \frac{\dot{\alpha}_t}{\alpha_t}x + \left(\frac{\dot{\alpha}_t\sigma_t^2}{\alpha_t} - \dot{\sigma}_t\sigma_t\right)\nabla \log \rho_t(x). \tag{G.5}$$

This proves the score/velocity relation used in Section 7.

H Small-noise field bridges

Consider $\mathrm{d}Y_s = \boldsymbol{F}(Y_s)\mathrm{d}s + \sqrt{2\varepsilon}\mathrm{d}B_s$ conditioned on $Y_0 = z, Y_1 = x$. Under a Freidlin–Wentzell large-deviation principle and suitable boundary regularity, paths have rate functional

$$I_{z,x}(\gamma) = \frac{1}{4}\int_0^1 \|\dot{\gamma}_s - \boldsymbol{F}(\gamma_s)\|^2\mathrm{d}s, \qquad \gamma(0) = z,\ \gamma(1) = x. \tag{H.1}$$

Therefore the Doob bridge concentrates, as $\varepsilon \to 0$, on minimizers of $I_{z,x}$. If $x = T_\Phi(z)$ and the deterministic field line is unique under the chosen clock, the minimizer has zero action after time reparameterization and the regularized bridge recovers the deterministic field-line bridge. This justifies treating Eq. (6.2) as the zero-temperature limit of field-biased stochastic bridges. A complete numerical algorithm for these bridges is outside the scope of this paper.

I Estimator and experimental details

Algorithmically, the Markovization diagnostic uses sampled bridge triples (Z_i, X_i, U_i) and a time grid. At each time t, compute $X_{t,i}$, fit a nearest-neighbor structure on $\{X_{t,i}\}_{i=1}^n$, estimate the conditional mean by averaging U_j over the neighbors of $X_{t,i}$, and average squared residuals. This is a nonparametric plug-in estimator of $\mathbb{E}\,\mathrm{Tr}\,\mathrm{Var}(U \mid X_t)$.

The main toy experiment uses only synthetic data. The target is an equally weighted mixture of eight Gaussians with centers $3(\cos(2\pi k/8), \sin(2\pi k/8))$ and component standard deviation 0.25. The base is $\mathcal{N}(0, I_2)$. The OT coupling is the Hungarian assignment minimizing squared Euclidean cost on the sampled minibatch. The conditional variance estimator uses $k = 32$ nearest neighbors in the bridge state X_t to estimate $\mathbb{E}[U \mid X_t]$ for $U = X - Z$. The 19 time points are equally spaced in $[0.05, 0.95]$. The replication figure uses seeds $0, 1, 2, 3, 4$ and sample sizes $n = 500, 1000$. The script `scripts/toy_projection_gap_replicates.py` regenerates `figures/toy_gap_replicates.csv` and the right panel of Figure 2.

J Closed-form gap for straight Gaussian bridges

The Markovization gap can be computed exactly in a simple case, which clarifies why endpoint coupling is not a cosmetic choice. Let $Z \sim \mathcal{N}(0, \Sigma_0)$, $X \sim \mathcal{N}(0, \Sigma_1)$, independently, and consider the straight bridge

$$S_t = (1-t)Z + tX, \qquad U = X - Z. \tag{J.1}$$

Because (U, S_t) is jointly Gaussian, the conditional variance is deterministic:

$$\begin{aligned}\mathrm{Var}(U \mid S_t) = \Sigma_0 + \Sigma_1 \\ - \big(t\Sigma_1 - (1-t)\Sigma_0\big)\big((1-t)^2\Sigma_0 + t^2\Sigma_1\big)^{-1}\big(t\Sigma_1 - (1-t)\Sigma_0\big).\end{aligned} \tag{J.2}$$

Therefore the instantaneous Markovization gap is

$$g(t) = \mathrm{Tr}\ \mathrm{Var}(U \mid S_t). \tag{J.3}$$

In the isotropic case $\Sigma_0 = \sigma_0^2 I_d$, $\Sigma_1 = \sigma_1^2 I_d$, Eq. (J.2) reduces to

$$g(t) = d\,\frac{\sigma_0^2\sigma_1^2}{(1-t)^2\sigma_0^2 + t^2\sigma_1^2}. \tag{J.4}$$

For $\sigma_0 = \sigma_1 = 1$, the integrated gap is

$$\int_0^1 g(t)\mathrm{d}t = d\int_0^1 \frac{\mathrm{d}t}{(1-t)^2 + t^2} = \frac{\pi d}{2}. \tag{J.5}$$

Thus even a perfectly Gaussian independent straight bridge has a strictly positive irreducible compression cost. This is not an architecture problem: it arises because many endpoint pairs pass through the same intermediate state with different velocities. In contrast, a deterministic nonintersecting Monge bridge has zero gap.

Proposition 2 (Zero gap for nonintersecting deterministic bridges)**.** *Let $X = T(Z)$ be a deterministic endpoint coupling and let $\gamma_t(z)$ be a deterministic bridge with velocity $u_t(z) = \partial_t\gamma_t(z)$. If $z \mapsto \gamma_t(z)$ is injective π_0-a.s. for almost every t, then $\mathfrak{G}(\mathsf{Q}) = 0$.*

Proof. When $z \mapsto \gamma_t(z)$ is injective, the current state $Y = \gamma_t(Z)$ determines Z outside a null set. Hence $u_t(Z)$ is measurable with respect to Y, so $\mathrm{Var}(u_t(Z) \mid \gamma_t(Z)) = 0$. Integrating over time gives $\mathfrak{G}(\mathsf{Q}) = 0$. □

This proposition gives a precise version of the intuition behind rectification and OT couplings. A bridge is easy to Markovize when intermediate states retain enough information to identify the hidden endpoint pairing. Crossings, caustics, and independent pairings destroy this information and create conditional velocity variance.

K BGM factorization and conditional independences

A Bridge Graphical Model can be represented as a continuous-depth latent graphical model. In a discrete approximation with times $0 = t_0 < t_1 < \cdots < t_K = 1$, a generic inference-side factorization is

$$q_{\phi,\lambda}(z, x_{1:K-1} \mid x_K) = q_\phi(z \mid x_K) \prod_{k=1}^{K-1} r_\lambda(x_k \mid x_{k-1}, x_K, z), \tag{K.1}$$

where the bridge factors are allowed to be non-causal because they condition on the endpoint x_K. The decoder factorization is causal:

$$p_\theta(x_{0:K}) = p_0(x_0) \prod_{k=1}^{K} p_\theta(x_k \mid x_{k-1}). \tag{K.2}$$

The Markov projection is the operation that replaces the non-causal local transition information in Eq. (K.1) with a causal transition depending only on (x_{k-1}, t_{k-1}). In the continuous-time limit, this becomes the conditional expectation $v^\star(x, t) = \mathbb{E}[u_t \mid X_t = x]$.

This factorization also explains why the endpoint coupling is an inference object. If $q_\phi(z \mid x)$ is learned, then the coupling $\Gamma_\phi(\mathrm{d}z, \mathrm{d}x) = q_\phi(\mathrm{d}z \mid x)\pi_{\mathrm{data}}(\mathrm{d}x)$ plays exactly the role of an encoder distribution. Fixed independent noise corresponds to an uninformative encoder; OT coupling corresponds to a deterministic or low-entropy encoder obtained by solving a transport problem; stochastic encoders interpolate between these extremes. Diffusion, flow matching, and field matching differ less in the graphical model than in which factors are fixed, optimized, or projected.

L Additional diagnostic results

The nearest-neighbor estimator introduces a smoothing parameter k. Table 4 reports a small sensitivity check on the same eight-Gaussian task with $n = 1000$ and five seeds. The absolute value of the OT estimate increases with k, as expected from oversmoothing local conditional means, but the qualitative conclusion is stable: OT coupling remains roughly two orders of magnitude easier to Markovize than independent coupling.

Table 4: Sensitivity of integrated Markovization gap to the nearest-neighbor parameter k, using $n = 1000$ and seeds $0, \ldots, 4$.

k	independent coupling	minibatch OT coupling	reduction factor
16	4.803 ± 0.111	0.0278 ± 0.0022	$172.8\times$
32	4.977 ± 0.109	0.0474 ± 0.0033	$104.9\times$
64	5.130 ± 0.110	0.0864 ± 0.0047	$59.3\times$

A useful way to read these numbers is not as a final estimator of the true population gap, but as a bridge-design score. The same estimator, applied to the same bridge family and data, can rank couplings before neural training. The diagnostic is therefore analogous to a preconditioner: it does not solve the generative problem, but it identifies endpoint/bridge choices that will impose a lower irreducible regression floor on the learner.

M Anonymous release and reproducibility checklist for the code

The supplementary code should expose three entry points: (i) generation of the single-run curve in Figure 2; (ii) replication over seeds and sample sizes; and (iii) the nearest-neighbor sensitivity table above. For the current source package, the replication script is

`scripts/toy_projection_gap_replicates.py`.

The synthetic experiment has no learned parameters and requires only NumPy, SciPy, scikit-learn, and Matplotlib. Its main randomness comes from endpoint sampling and target-component sampling. The OT coupling uses the exact Hungarian assignment on each minibatch; no Sinkhorn regularization

or stochastic optimizer is used. This makes the diagnostic reproducible enough to be checked by reviewers in seconds to minutes, depending on sample size.

For a NeurIPS anonymous submission, the repository should avoid institution-specific paths, author names, and non-anonymous commit history. The arXiv/JMLR version can retain the named author metadata, but the conference supplementary ZIP should use anonymous filenames and a README that states how to reproduce each figure without linking to a public author-identifying repository during the review period.

N What would make the framework a full model class?

The present paper mostly fixes or externally chooses Γ and R. A fully learnable Bridge Graphical Model would optimize

$$\max_{\phi,\lambda,\theta,\psi} \mathbb{E}_{\pi_{\mathrm{data}}(x)} \mathbb{E}_{\mathsf{Q}_{\phi,\lambda}(\cdot|x)} \left[\log p_\psi(x \mid Y_1) - \frac{\mathrm{d}\mathsf{Q}_{\phi,\lambda}(\cdot \mid x)}{\mathrm{d}\mathsf{P}_\theta}(Y_{0:1}) \right], \tag{N.1}$$

with the Radon–Nikodym term interpreted through Girsanov in stochastic gauges or through matching/optimal-control relaxations in deterministic gauges. Three computational difficulties then appear. First, sampling from $\mathsf{R}_\lambda^{z,x}$ must remain cheap enough for minibatch training. Second, the local target u_t or β_t must be computable or estimable without solving a global bridge problem at every step. Third, the gauge a_t should be chosen for numerical stability while preserving the learned current.

These difficulties are also opportunities. Learning Γ_ϕ imports representation learning into diffusion/flow training; learning R_λ imports path-posterior design; learning or adapting the gauge imports sampler design. The BGM decomposition makes these three choices explicit and therefore makes it possible to ablate them independently.

O Technical comparison with closest unification frameworks

Stochastic interpolants specify an interpolating random variable, often written abstractly as $I_t(x_0, x_1, \xi)$, and then learn the velocity or score associated with the induced marginal path. In BGM language, this corresponds to choosing a coupling Γ and a bridge kernel $\mathsf{R}^{z,x}$ whose samples are $X_t = I_t(z, x, \xi)$. The BGM decomposition is therefore compatible with stochastic interpolants, but it makes two additional degrees of freedom explicit: the endpoint coupling can be learned as an encoder, and the same marginal path can later be represented in different gauges. This distinction matters when one wants to separate representation learning from sampler design.

Diffusion Flow Matching is even closer: it explicitly studies coupling, deterministic or stochastic bridges, and Markovian projection. The extra object introduced here is the Markovization gap as a scalar diagnostic of how lossy that projection is. The projection identity alone says what the optimal velocity is; the gap says how much non-causal bridge information cannot be recovered by any Markov vector field. This is the quantity estimated in the toy experiment and analyzed in closed form in Appendix J.

Latent stochastic interpolants add an encoder and decoder around stochastic interpolants. BGM adopts the same variational intuition but separates the bridge kernel from the gauge. This separation is useful because a learned latent path posterior may be sampled as a deterministic ODE, a stochastic SDE, or an augmented field without changing the endpoint coupling. In other words, the encoder answers "which latent endpoint and bridge path explain this datapoint?", while the gauge answers "how should the same current be represented numerically?"

Field and interaction matching approaches begin from a different primitive: an augmented potential or interaction field rather than a time-indexed interpolation. BGM imports these models by constructing a hitting map and field-line bridge kernel. This is not simply a notational change. It gives field methods an endpoint coupling, a bridge law, and a Markovization gap, making them comparable to diffusion and flow methods with the same diagnostic. The caustic gap is the field analogue of the projection loss: it measures when an augmented field line carries endpoint information that collapses under a Markov velocity on the observed space.

P Potential-field bridge details

A Poisson-style field bridge can be described in terms of a signed or nonnegative source measure ν on an augmented domain $\Omega \subset \mathbb{R}^{d+D}$. A potential Φ solves, in the weak sense,

$$-\Delta\Phi = c_{d,D}\nu, \tag{P.1}$$

with boundary conditions chosen by the model class. The field is $\boldsymbol{F} = -\nabla\Phi$. If S_0 is a simple source surface and S_1 is the data surface, then field-line generation starts from $Z \sim \mu_0$ on S_0, follows $\dot{y}_s = \boldsymbol{F}(y_s)$, and stops when the trajectory hits S_1. The hitting map $T_\Phi : S_0 \to S_1$ then induces the coupling $(\mathrm{Id}, T_\Phi)_\# \mu_0$.

The time parameter used by a generative model is not necessarily the physical field-line time s. A clock $r_z : [0,1] \to [0, \tau(z)]$ creates the bridge path $\gamma_z(t) = \varphi_{r_z(t)}(z)$. Different clocks preserve the same geometric field lines but change the velocity target $u_t(z) = r_z'(t)\boldsymbol{F}(\gamma_z(t))$ and therefore change the Markovization gap. This is another bridge-design degree of freedom: two models can share the same potential field but induce different regression problems because they traverse the field at different speeds.

Singularities are unavoidable when data are represented as charges. The deterministic bridge is well-defined only away from singular sets and only for trajectories that hit the data surface transversally. The small-noise Doob bridge in Appendix H regularizes this by replacing deterministic field lines with field-biased stochastic paths conditioned on endpoints. This yields a continuum between an electrostatic bridge and an entropy-regularized bridge, and it suggests a practical path for training: use small noise to avoid singular caustics early in training, then anneal toward sharper field-line transport.

Q Real-data latent benchmark details

The real-data benchmark uses the same BGM components as the synthetic diagnostic but replaces the target distribution with real-image latents. Images are flattened, standardized, embedded with PCA whitening, and paired with standard normal base samples. The benchmark reports (i) integrated Markovization gap, (ii) velocity-regression error for a fixed ridge model, and (iii) a latent two-sample metric. The exact commands used for the LFW subset are:

```
PYTHONPATH=src python -B scripts/run_realdata_gap.py \
  --dataset lfw --n 150 --seeds 0 1 2 --k-neighbors 16
PYTHONPATH=src python -B scripts/run_realdata_benchmark.py \
  --dataset lfw --n 150 --seeds 0 1 2 --k-neighbors 16 --ridge-repeats 8
```

The same scripts accept `-dataset mnist`, `-dataset fashionmnist`, and `-dataset cifar10`; those runs require local torchvision data download. The benchmark is deliberately latent and low-compute. It evaluates the BGM diagnostic, not image-generation state of the art.

R Ethics, existing assets, new assets, and compute

Existing assets. The reported real-data experiment uses the LFW face subset [5] distributed through scikit-image [22], with images embedded only into low-dimensional PCA latents for a diagnostic regression task; it is not used for identification, recognition, or deployment. The software dependencies include NumPy, SciPy, scikit-learn [14], scikit-image [22], Matplotlib, and optionally torchvision for local MNIST [10], Fashion-MNIST [23], and CIFAR-10 [9] latent runs. Because LFW contains public web photographs whose image copyrights remain with original rights holders, the source package does not redistribute raw LFW images. It also does not redistribute raw MNIST, Fashion-MNIST, or CIFAR-10 datasets; optional scripts download them through the user's local environment when explicitly requested. The released code and generated diagnostic outputs are provided under the repository license.

New assets. The new assets are the anonymized source code, LaTeX source, generated diagnostic figures, and CSV summaries. The released code is intended to reproduce the Markovization-gap and latent-regression diagnostics only; no trained image generator, large model checkpoint, or new dataset is released.

Compute. All reported experiments are CPU-only. The synthetic diagnostic runs in seconds to minutes on a standard laptop-class CPU; the LFW latent benchmark uses $n = 150$, three seeds, a 16-dimensional PCA embedding, nearest-neighbor conditional-variance estimates, and a closed-form ridge model, and also runs without a GPU.

Broader impacts and safeguards. The positive impact is conceptual and methodological: the framework may help researchers compare bridge, coupling, and gauge choices before expensive generative training. The main negative impact is indirect: improved generative modeling methods can contribute to dual-use image generation, including misleading synthetic media, if later scaled into high-fidelity generators. This paper does not release a deployable generator or pretrained model; the reproducible release is limited to diagnostics, proof code, and low-dimensional latent experiments.

S Suggested NeurIPS ablation plan

The present submission is intentionally concept-and-feasibility, but the framework suggests a direct ablation ladder. The first ablation fixes the bridge to be straight and varies only the coupling: independent, minibatch OT, entropic OT, and encoder-induced coupling. The second fixes the coupling and varies the bridge: straight deterministic, Brownian, small-noise field-biased, and Schrödinger-like. The third fixes the learned current and varies the gauge: ODE, isotropic SDE with score correction, and augmented field. The fourth learns $q_\phi(z \mid x)$ jointly with a latent bridge, testing whether the endpoint encoder lowers the estimated Markovization gap and improves downstream sample quality.

The key experimental prediction is not merely that OT or learned couplings produce nicer paths. The prediction is that lower estimated Markovization gap should correlate with lower training loss floor, faster convergence, and fewer solver steps for comparable sample quality. This is falsifiable: if two bridge choices have very different $\mathfrak{G}$ but identical optimization behavior and sample quality under controlled architectures, then the diagnostic is incomplete. Conversely, if $\mathfrak{G}$ predicts optimization difficulty across datasets and architectures, it becomes a useful model-selection tool before expensive generative training.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes] , [No] , or [N/A] .
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No] , it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., "error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Introduction state that this is a concept-and-feasibility framework with proof-level diagnostics and minimal empirical tests, not a full image-generation benchmark. The main claims are supported by Sections 3–8 and qualified in Section 10.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

Justification: Section 10 discusses the limited empirical scale, assumptions in the stability theorem, high-dimensional estimation issues, and field-line regularity assumptions. Appendices S and R further delimit what the current experiments and release do not claim.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The theorem and proposition statements in Sections 4, 5, and 6 state their assumptions, with proof details in Appendices A–H. Appendix D gives the stability proof and Appendix J gives the closed-form Gaussian gap results.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 8 describes the synthetic and LFW-latent experiments, while Appendices I, L, and Q give distributions, sample sizes, seeds, estimators, commands, and evaluation metrics. Appendix M states the scripts needed to reproduce figures and tables.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer:[Yes]

Justification: Appendices M and Q provide exact commands for reproducing the diagnostic results, and the supplementary repository contains the scripts and generated CSV/PDF outputs. Appendix R states that raw external datasets are not redistributed; optional real-data scripts download data locally when explicitly requested.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.

- Please see the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 8 specifies the bridge, couplings, latent dimension, PCA whitening, fixed ridge model, and reported metrics. Appendices I, Q, and L give the time grid, nearest-neighbor estimator, seeds, sample sizes, and sensitivity settings.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Figure 2 reports five-seed synthetic replications, and Table 3 reports mean $\pm$ standard deviation over three LFW-latent seeds. Appendix L gives sensitivity results for the nearest-neighbor estimator.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).

- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix R states that all reported experiments are CPU-only, gives the LFW-latent configuration, and notes that no GPU is required. Appendix M describes the synthetic diagnostic as a seconds-to-minutes computation depending on sample size.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The work is a theoretical and diagnostic study, does not train or release a deployable generator, and does not conduct new human-subject or crowdsourcing studies. Appendix R describes the use of public image assets, non-identification latent processing, and the absence of model deployment.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No] , they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Appendix R discusses the positive methodological impact of more interpretable and efficient bridge selection, and the indirect negative dual-use risk that improved generative modeling can contribute to misleading synthetic media if later scaled. Section 10 also limits the current claims to diagnostics rather than deployment.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No] , they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: Appendix R states that the release does not include a trained image generator, large model checkpoint, or new scraped dataset. The released assets are diagnostic code, generated CSVs/figures, and LaTeX source, so high-risk model-release safeguards are not applicable.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: Appendix R credits the LFW subset and software packages, and the source package does not redistribute raw datasets. We answer [No] rather than [Yes] because the copyright status of individual LFW web photographs is not fully explicit; this should be verified or the default benchmark should be switched to a clearer-license dataset before final submission.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: Appendices M and R document the new code, generated figures, generated CSVs, intended use, and release scope. The supplementary README gives compile and rerun commands, and no new dataset or trained model checkpoint is introduced.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not conduct crowdsourcing, recruit participants, or collect new human-subject data. Sections 8 and Appendix Q use synthetic data and a public LFW subset only for a non-identification latent diagnostic.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not recruit participants, interact with people, or collect new human-subject data; Appendix R describes the public-data-only release scope. Therefore IRB approval or equivalent review is not applicable to the reported experiments.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: No LLM is used as part of the model, theorem, experiment, data pipeline, or evaluation method; the method is described in Sections 3–6 and the experiments in Section 8. If authors used language assistance only for writing, editing, or formatting, that is outside the core methodology under the checklist wording.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.